

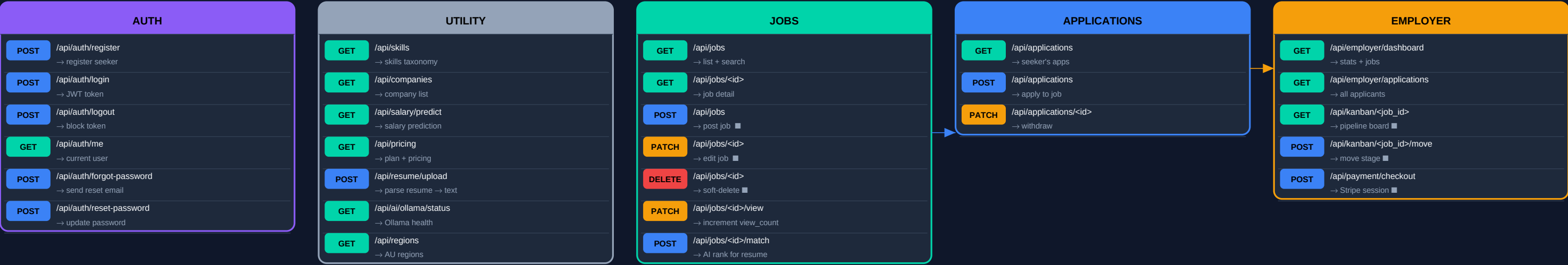
OpenJobs — System Architecture

Flask API · SQLite · Ollama (gemma3:4b) · Telegram Bot · Stripe · SMTP



API Routes — 40+ Endpoints

Bold = auth-gated · AUTH / UTILITY → JOBS → APPLICATIONS → EMPLOYER → AI → CRM / ADMIN



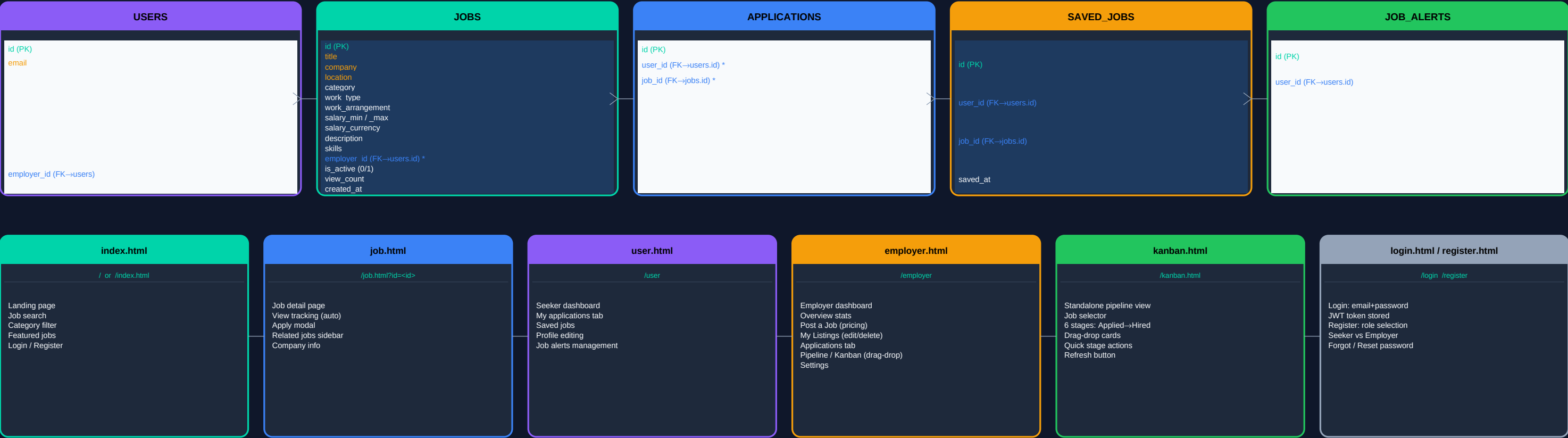
AI LAYER (semantic_matcher.py + ai_matcher.py)

Hybrid scoring: keyword-first (fast) → Ollama semantic rescore (borderline 1-3 keyword matches only)

Models: gemma3:4b · 45s timeout · Smart threshold: skip Ollama if keyword score ≥ 4 or = 0

Data Models & Frontend Pages

Primary keys + foreign keys shown · Frontend routes to templates



KEY INSIGHT: jobs.employer_id → users.id (FK) applications.job_id → jobs.id (FK) applications.user_id → users.id (FK)

Ollama only fires for borderline keyword scores (1-3). Clear matches skip AI → saves ~70% of LLM calls. Employer posts job → email_alert → matching seekers notified.